

THREAD POOL (EXECUTOR FRAMEWORK)

(1) EXPLANATION OF THREAD POOL?

Creating a new thread for every job , may create performance on memory problem. To over come this problem we should go for thread pool.

Thread pool is pool of already created thread , ready to our job

Java 1.5 version introduced thread pool framework to implement thread pools

Thread pool framework also known as executor framework

We can create a thread pool as followed :

```
ExecutorService service = new Executor.newFixedThreadPool(3);
```

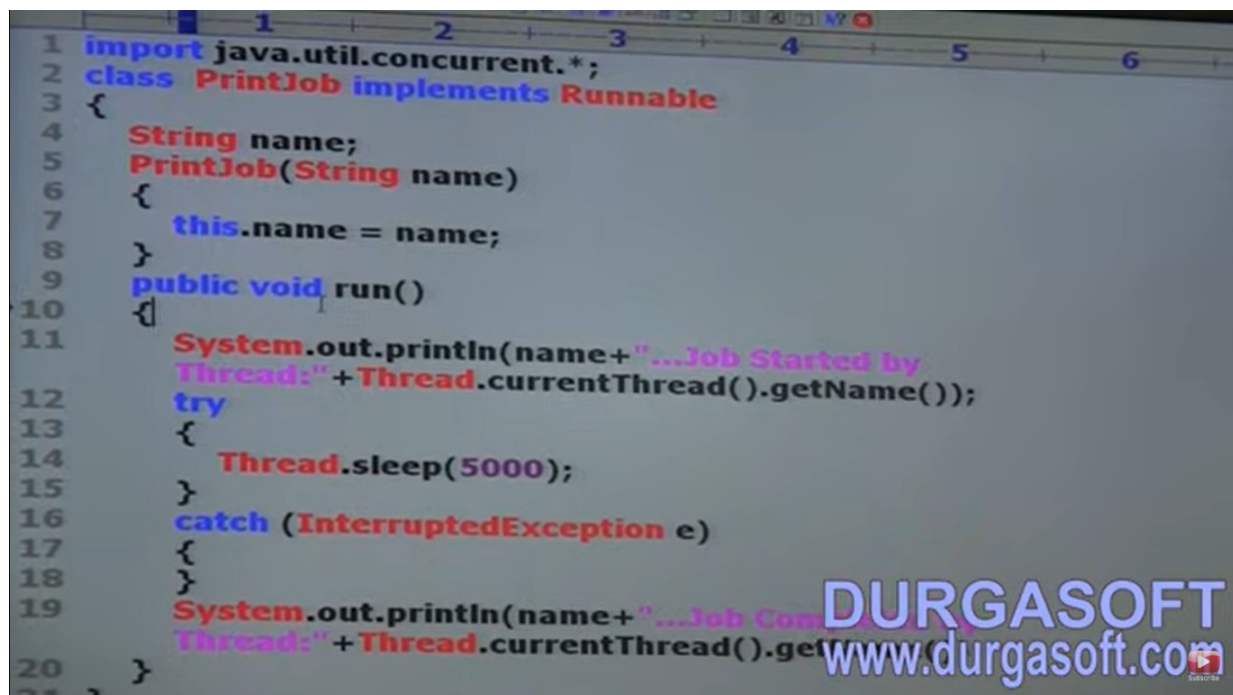
We can submit a runnable job by using submit method

```
Service.submit(job);
```

We can shutdown executor service by using shutdown method

```
Service.shutdown()
```

Ex-



```
1 import java.util.concurrent.*;
2 class PrintJob implements Runnable
3 {
4     String name;
5     PrintJob(String name)
6     {
7         this.name = name;
8     }
9     public void run()
10    {
11        System.out.println(name+"...Job Started by
12        Thread:"+Thread.currentThread().getName());
13        try
14        {
15            Thread.sleep(5000);
16        }
17        catch (InterruptedException e)
18        {
19        }
20        System.out.println(name+"...Job Completed by
21        Thread:"+Thread.currentThread().getName());
22    }
23 }
```

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```
22 class ExecutorDemo
23 {
24     public static void main(String[] args)
25     {
26         PrintJob[] jobs={ new PrintJob("durga"),
27                             new PrintJob("Ravi"),
28                             new PrintJob("shiva"),
29                             new PrintJob("pavan"),
30                             new PrintJob("Suresh"),
31                             new PrintJob("Anil") };
32         ExecutorService service= Executors.newFixedThreadPool(3);
33         for(PrintJob job: jobs)
34         {
35             service.submit(job);
36         }
37         service.shutdown();
38     }
39 }
40 }
41 }
```

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